

NETWORK ROUTING AND SWITCHING

COURSE PROJECT PROPOSAL GUIDELINES

Advanced & Complex Enterprise Network Project

Project Title

Design, Implementation, Security, and Performance Evaluation of a Scalable Multi-Site Enterprise Network Using Advanced Routing, Switching, Redundancy, and Network Security Technologies

1. Project Overview

This project requires students to design, implement, secure, test, and evaluate a large-scale enterprise network representing a realistic organization with multiple departments, buildings, branch offices, a data center, disaster-recovery facilities, guest users, wireless users, voice services, and Internet connectivity.

- Multiple sites and departments
- VLAN segmentation and inter-VLAN routing
- IPv4 and IPv6 dual-stack operation
- Multi-area OSPF and route summarization
- DHCP, DNS, Web and other network services
- NAT/PAT and ACL-based security
- STP/RSTP, EtherChannel and gateway redundancy
- WAN connectivity and QoS
- Network monitoring concepts
- Failure injection, recovery and performance evaluation

2. Project Scenario

Assume that a large organization operates a Head Office, two Branch Offices, a Data Center, and a Disaster Recovery site. The existing network is flat, difficult to manage, poorly secured, and lacks redundancy and scalability. Students must redesign the network using enterprise networking principles.

- Head Office
- Branch Office 1
- Branch Office 2
- Data Center
- Disaster Recovery Site
- Multiple departments
- Internal and public-facing servers
- Guest, voice, wireless and IoT networks

3. Main Project Aim

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To design, configure, secure, test, and evaluate a scalable, redundant, secure, and high-performance enterprise network using advanced routing and switching technologies.

4. Project Objectives

1. Design a hierarchical enterprise network using Core, Distribution, and Access layers.
2. Develop an efficient IPv4 addressing and VLSM plan.
3. Implement IPv6 dual-stack networking.
4. Configure multiple VLANs and 802.1Q trunking.
5. Implement Layer-3 inter-VLAN routing.
6. Configure multi-area OSPF, route summarization, and default routing.
7. Implement STP/RSTP and EtherChannel.
8. Implement first-hop gateway redundancy such as HSRP, VRRP, or GLBP where supported.
9. Configure DHCP, DHCP relay, DNS, and Web services.
10. Implement NAT/PAT and appropriate ACL policies.
11. Apply Layer-2 security including Port Security, BPDU Guard and secure unused ports.
12. Configure secure device management using SSH.
13. Design WAN and Internet connectivity.
14. Develop a DMZ for public-facing services.
15. Apply QoS concepts for voice and video traffic.
16. Conduct failure, recovery, convergence, and performance testing.
17. Produce professional network documentation and defend the design in a presentation and viva.

5. Proposed Network Architecture

Site	Minimum Recommended Devices	Main Functions
Head Office	2 Core/Distribution switches or routers; 4 Access switches	Departments, VLANs, users, voice, wireless
Branch 1	1 router; 2 switches	Branch departments and users
Branch 2	1 router; 2 switches	Branch departments and users
Data Center	2 routers; 2 distribution switches	Servers, DMZ, internal services
Disaster Recovery	1 router; 2 switches	Backup and recovery services

6. Hierarchical Network Design

The proposed architecture should follow the three-layer enterprise model: Core, Distribution, and Access. Redundant paths should be provided where practical.

Internet → Edge Routers → Core Layer → Distribution Layer → Access Switches →
Users/Servers/VoIP/Wireless

7. VLAN Architecture

VLAN	Name	Purpose
10	ADMIN	Administration
20	HR	Human Resources
30	FINANCE	Finance
40	IT	IT Department
50	ENGINEERING	Engineering

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60	USER	General Users
70	SERVER	Internal Servers
80	VOICE	VoIP
90	GUEST	Guest Network
100	MANAGEMENT	Network Management
110	SECURITY	Security Systems
120	IoT	IoT Devices

8. IPv4 Addressing and VLSM

Use a structured enterprise address block, such as 10.0.0.0/8, and divide it using VLSM and hierarchical summarization.

Location	Example Address Block
Head Office	10.10.0.0/16
Branch 1	10.20.0.0/16
Branch 2	10.30.0.0/16
Data Center	10.40.0.0/16
Disaster Recovery	10.50.0.0/16

9. IPv6 Implementation

The project should support IPv4/IPv6 dual stack. Students should configure IPv6 addressing, gateways, routing, and IPv6 ACLs. Example documentation prefix: 2001:DB8:10::/48.

10. Advanced Routing Design

OSPF should be the primary dynamic routing protocol. A suggested multi-area architecture is:

Area	Purpose
Area 0	Core backbone
Area 10	Head Office
Area 20	Branch Office 1
Area 30	Branch Office 2
Area 40	Data Center

- Router ID configuration
- OSPF neighbor verification
- Passive interfaces
- Cost manipulation
- Default-route advertisement
- Route summarization
- OSPF authentication
- Stub/NSSA concepts where appropriate
- Load balancing and convergence testing

11. Inter-VLAN Routing

Implement inter-VLAN routing using Layer-3 switching where possible. Router-on-a-stick may be used as an alternative for selected segments. Verify SVI status, gateway reachability, and end-to-end communication.

12. Advanced Switching

- 802.1Q trunking
- STP/RSTP root and backup-root design
- PortFast and BPDU Guard
- Root Guard where appropriate
- EtherChannel using LACP or PAgP
- Layer-3 switching
- Redundant uplinks

13. First-Hop Redundancy

Implement HSRP, VRRP, or GLBP where supported. Configure a virtual default gateway so that end users can continue communication if the primary gateway fails.

14. WAN Architecture

Connect branch sites to the Head Office using a simulated WAN. Students may use point-to-point links, Ethernet WAN, GRE/IPsec concepts, MPLS concepts, or other technologies available in their simulator. The selected design must be justified.

15. Internet Connectivity and NAT

- Default route
- NAT/PAT
- Inside/outside interfaces
- Private-to-public translation
- Internet reachability testing

16. Network Security Architecture

- Port Security
- BPDU Guard
- PortFast
- Shutdown of unused ports
- Standard ACLs
- Extended ACLs
- IPv6 ACLs
- SSH management
- Local authentication
- Management VLAN
- Guest network isolation

17. Required ACL Policies

18. Guest users must not access internal server networks.
19. General users must not access the Finance VLAN unless explicitly permitted.
20. Only authorized IT administrators may manage network devices remotely.

21. Public-facing services should be isolated in a DMZ.
22. Required voice and business traffic must remain permitted.
23. All ACL rules must be documented and tested.

18. DHCP and Network Services

- DHCP scopes for multiple VLANs
- DHCP relay
- DNS service
- Web service
- Email/file/database services where supported
- DMZ for public-facing services

19. Data Center and DMZ Design

The Data Center should contain internal and public-facing services. Public services should be separated from internal systems through a DMZ/security boundary.

Internet → Edge Router → Security Boundary → DMZ (Web/DNS) and Internal Server Network

20. QoS Design

Traffic Type	Suggested Priority
VoIP	Very High
Video Conference	High
Business Applications	Medium
Web Browsing	Normal
Guest Traffic	Low

Students should explain how QoS policies can reduce delay, jitter, and packet loss for time-sensitive traffic.

21. Network Monitoring

- Syslog concepts
- SNMP concepts
- Interface monitoring
- ICMP monitoring
- NetFlow concepts where supported
- Latency, packet loss and utilization measurements

22. Mandatory Failure and Recovery Testing

Test	Failure Scenario	Required Observation
1	Disconnect a core link	Path change and recovery
2	Shutdown a routing interface	OSPF reconvergence
3	Disable the primary gateway	Gateway failover
4	Disable one EtherChannel member	Bundle resilience
5	Shutdown an OSPF router	Route reconvergence
6	Block a trunk	VLAN/path recovery

7	Disable DHCP service	Impact and mitigation
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For every test, document the initial condition, failure introduced, observed behavior, convergence/recovery behavior, and final condition. Do not invent performance values; report measurements from actual testing.

23. Performance Evaluation

Evaluate the implemented network using measurable indicators such as latency, packet loss, throughput, routing convergence, interface utilization, and availability. Compare alternative designs or configurations where possible.

24. Required Project Deliverables

24. Project proposal of approximately 8–12 pages.
25. Logical and physical network topology diagrams.
26. Working Packet Tracer, GNS3, or EVE-NG project file.
27. Complete router and switch configuration files.
28. IPv4/VLSM and IPv6 addressing tables.
29. VLAN and trunking documentation.
30. Routing and OSPF documentation.
31. Security and ACL policy document.
32. Testing and troubleshooting report.
33. Failure/recovery analysis.
34. Performance evaluation.
35. Final technical report of approximately 40–60 pages.
36. 15–20 minute presentation.
37. Individual viva.

25. Final Report Structure

- Chapter 1 — Introduction
- Chapter 2 — Network Requirements
- Chapter 3 — Network Architecture
- Chapter 4 — IP and VLAN Design
- Chapter 5 — Routing Design
- Chapter 6 — Switching Design
- Chapter 7 — Network Security
- Chapter 8 — Network Services
- Chapter 9 — Implementation
- Chapter 10 — Testing and Evaluation
- Chapter 11 — Discussion
- Chapter 12 — Conclusion and Future Work
- References and Appendices

26. Suggested Evidence and Verification Commands

```
show running-config
show ip interface brief
show vlan brief
show interfaces trunk
```

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```
show etherchannel summary
show spanning-tree
show ip route
show ip ospf neighbor
show ip ospf interface
show ip protocols
show access-lists
show standby
show ipv6 interface brief
show ipv6 route
```

27. Advanced Assessment Rubric

Component	Marks
Project Proposal	5
Network Architecture	10
IP Addressing & VLSM	10
VLAN & Switching	10
OSPF/Routing	15
IPv6	5
Network Security	10
WAN & NAT	5
Network Services	5
QoS & Redundancy	5
Failure/Recovery Testing	5
Performance Evaluation	5
Final Report	5
Presentation & Viva	5
Total	100

28. Minimum Advanced Requirements

- Minimum 3 network sites
- Minimum 8 VLANs
- VLSM
- IPv4 and IPv6
- Multi-area OSPF
- Route summarization
- Inter-VLAN routing
- 802.1Q trunking
- STP/RSTP
- EtherChannel
- First-hop redundancy
- DHCP and DHCP relay
- DNS and Web service
- NAT/PAT
- Standard and extended ACLs
- SSH
- Port security
- WAN connectivity
- DMZ
- QoS design

- Network monitoring concept
- Failure testing
- Recovery testing
- Performance evaluation

29. Recommended Highest-Level Project

Design and Implementation of a Highly Available Multi-Site Enterprise Network with Multi-Area OSPF, IPv6 Dual Stack, VLAN Segmentation, Layer-3 Switching, EtherChannel, STP, HSRP/VRRP, NAT, ACL, DHCP, DMZ, QoS, and Automated Failure-Recovery Analysis.

30. Evaluation Philosophy

The project should be evaluated as a complete network-engineering exercise rather than simply a working Packet Tracer topology. Students must be able to explain why each technology was selected, how it was configured, what problem it solves, and how its behavior was verified.

Requirement Analysis → Architecture → IP Planning → VLAN Design → Routing Design → Security Design → Implementation → Verification → Failure Injection → Recovery → Performance Analysis → Documentation → Presentation